# A Retained Mixed Gravity–Maxwell Bracket under Cyclic Homological Reduction through $\ell_3$
## Exact Transfer and Causal Cartan Compatibility

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**Abstract**

We transfer the classical gravity–Maxwell BV interaction on a compact positive Berger-clock background from a complete 64-row presentation to a typed cyclic 36-row retained complex. The contraction carries the exact odd pairing and the unary causal Green data. In the repository's suspended graded-symmetric convention the retained mixed ternary operation is nonzero. Two zero-derivative witnesses have exact coefficients $+1$ and $-1$ in the gravity-equation and Maxwell-equation sectors, respectively. Thus nonvanishing is witnessed by explicit evaluations rather than inferred from the total sparse-coefficient count.

The homological exchange channel is computed rather than omitted. It contains 342 canonical full-complex coefficients in the only nonempty channel, all supported on full row 38, and the certified retained projection annihilates that row. A one-entry projection mutation exposes all 342 terms. Thus the retained exchange contribution vanishes for a structural reason while the direct $q_3$ contact survives for this frozen contraction. Standard cyclic homotopy transfer supplies the abstract $L_\infty$ and cyclic identities; an independent sparse implementation audit replays them on all 36 rows. Lowering $\ell_3$ with the retained typed pairing independently reproduces all 25,662 degree-zero lowered quartic coefficients with zero cyclicity defects; changing the Maxwell pairing weight from two to one produces 17,108 exact defects on 14 rows. The remaining 288 ghost/antifield coefficients also replay exactly under the suspended-Darboux transpose, closing cyclicity of the full retained BV tensor.

For the explicitly specified cyclic strong deformation retract these calculations establish

$$\boxed{\text{The retained mixed gravity–Maxwell bracket survives this cyclic homological reduction through } \ell_3.}$$

The separately certified causal theorem gives compatible advanced and retarded unary homotopies and two-sided-causal-hull Cartan primitives for the background-fixing helical generator $K_{\mathrm{Berger}}$. It is a parallel compatibility theorem: $\ell_3$ is transferred with the support-local algebraic homotopy $S$, not by integrating out fields with a causal propagator. The statement is classical and carries separate LOCAL-ALGEBRAIC and LORENTZIAN-CAUSAL dependencies. It does not prove a nonzero operation on cohomology, invariance under changing the cyclic retract, or nonremovability under every unrestricted cyclic $L_\infty$ equivalence. Residual mixing, Hadamard products, QME restoration, and every quantum or particle interpretation remain open. The first deformation-filtration screen is now exact: after setting every PBW derivative word to zero, the complete 550-dimensional mixed $g^2A^2$ quartic sector is the image of admissible constant-field cyclic redefinitions, and a 51-coefficient $F_3$ primitive removes the landed 63-coordinate constant term. Thus the displayed

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zero-derivative evaluations witness representative nonvanishing but cannot witness deformation-class nontriviality. The complete first-jet physical page is also exact: retaining all zero-jet kernel freedom, the coupled Schur system admits an explicit four-axis positive-jet primitive. The complete full-BV filtered calculation has now returned the opposite invariant verdict. After coupling all zero-page kernel freedom, a normalized 22-row dual functional annihilates all 5,984 zero-page cyclic directions and all 14,998 first-jet directions on each axis, but evaluates to one on the first associated-graded residual. Hence the landed mixed bracket is not removable through total PBW order two by the declared nonnegative filtered, derivative-aware cyclic $F_2/F_3$ transformations. This is a nontrivial class in that scoped filtered deformation quotient, not yet a theorem in the unrestricted cyclic deformation complex or on residual cohomology. A separate exact audit obstructs the canonical local same-bundle branch projector on the 36-row carrier. The resulting rank-46 spatial-STF2 graph prolongation and its cyclic dual are now constructed as an exact local cyclic carrier. Its principal wave module is a nonsplit Einstein/extra-Weyl filtration, so the next branch gate is the exact subprincipal extension rather than a principal direct-sum projector.

## AI-use disclosure and computational accountability

OpenAI GPT-5 was used substantively for mathematical synthesis, verification code, claim auditing, and manuscript drafting. The human author directed the programme and remains responsible for the claims, citations, and final text. Every numerical count in the theorem is an exact sparse-coefficient count over $\mathbb{Q}(\sqrt{10})$ and is tied to a deterministic certificate. No floating point calculation enters an identity, rank, cyclicity test, or mutation witness. Submission remains conditional on final human review.

## Contents

## 1 Question and answer

The free BV complex identifies which classes survive gauge reduction, but it does not determine whether nonlinear gravity–matter vertices survive the same reduction. A transferred interaction can disappear in three distinct ways: its direct contact term can project to zero; homological exchange terms can cancel it; or cyclicity and the $L_\infty$ identities can fail after compression. Conversely, a nonzero tensor on a gauge-fixed carrier need not descend to any physical branch until its residual projection is computed.

This paper answers the algebraic question through ternary order for one explicit gravity–Maxwell extension and one frozen cyclic contraction of the positive Berger-clock complex. Write

$$\boxed{\begin{array}{c}\text{The retained mixed gravity–Maxwell bracket survives this}\\ \text{cyclic homological reduction through } \ell_3.\end{array}} \tag{1}$$

Here "survives" means that the explicitly transferred mixed ternary bracket is nonzero in the displayed retained coordinates and obeys the transferred arity-three identity. "Cyclic" means that the contraction is typed cyclic and that the complete lowered BV quartic form has been independently replayed under cyclic transposition. Causal Green and Cartan compatibility are proved separately in Section 6; they are not part of the definition of the algebraic transfer.

Nonvanishing in Theorem 4.1 is deliberately representative-level. By itself it does not prove

$$[\ell_3^{\text{mixed}}] \neq 0 \tag{2}$$

in a cyclic coderivation or Chevalley–Eilenberg deformation complex. Nor have we proved a nonzero induced operation on $H(\mathcal{R}_{36}, \ell_1)$, survival under all cyclic $L_\infty$ equivalences or further minimal transfer, or an obstruction to removing the selected component under every cyclic change of retained coordinates. Section 8 supplies one strictly scoped invariant strengthening: nonremovability in the declared nonnegative filtered, derivative-aware cyclic $F_2/F_3$ class through total PBW order two. It does not promote Equation (2) for the unrestricted deformation complex.

The theorem stops one rung before physical branch interpretation. A subsequent exact audit now obstructs the canonical support-local same-bundle Einstein-like/extra-Weyl projector on the 36-row carrier. Residual descent therefore requires an enlarged or differently organized carrier. The natural rank-46 STF2 graph carrier has now passed its exact unary cyclic-SDR gate. Its principal direct-sum anchor is obstructed, while the lower-order Berger splitting and nonlinear lift remain open, so a branch mixing table is still unauthorized. Deformation/vertex classes belong to a separate complex and are not used in the theorem below.

## 2 Complexes, conventions, and claim boundary

Let $(\mathcal{C}_{64}, q_1, \omega_{64})$ be the complete typed classical gravity–clock–Maxwell BV complex at the frozen rational Berger fixture. Its degree multiplicities are

$$6, 26, 26, 6 \qquad \text{in degrees } -1, 0, 1, 2. \tag{3}$$

Let $(\mathcal{R}_{36}, \ell_1, \omega_{36})$ denote the retained typed carrier, with degree multiplicities

$$4, 14, 14, 4. \tag{4}$$

The exact contraction is

$$(\mathcal{C}_{64}, q_1) \underset{\iota}{\overset{\pi}{\rightleftarrows}} (\mathcal{R}_{36}, \ell_1)\,, \qquad \pi\iota = \mathrm{id}, \qquad q_1 S + S q_1 = \mathrm{id} - \iota\pi, \tag{5}$$

with the exact side conditions

$$S^2 = 0, \qquad S\iota = 0, \qquad \pi S = 0, \tag{6}$$

and typed cyclic adjoint relations fixed by the carrier certificate. The coefficients lie in $\mathbb{Q}(\sqrt{10})$. The Maxwell field–equation pairing has absolute weight two, while the gravity weight is one. The signed odd-pairing ledger contains ten gravity entries of sign $-1$ and four Maxwell entries of sign $+2$ on the degree-zero field–equation pairs.

For completeness, let $\dagger$ denote the formal PBW adjoint, including integration by parts, and let $\Omega_{64}$ and $\Omega_{36}$ be the signed typed pairing matrices. All suspension and Koszul signs are already included in these matrices. The exact convention-specific cyclic SDR relations are

$$\iota^\dagger \Omega_{64} = \Omega_{36}\pi, \qquad \iota^\dagger \Omega_{64}\iota = \Omega_{36}, \qquad S^\dagger \Omega_{64} + \Omega_{64} S = 0. \tag{7}$$

Equivalently, for homogeneous retained $x$ and full-carrier $y$,

$$\omega_{64}(\iota x, y) = \omega_{36}(x, \pi y), \qquad \omega_{64}(Sx, y) = -\omega_{64}(x, Sy). \tag{8}$$

The last displayed sign is therefore the suspended-coordinate sign actually verified by the certificate, not an unsuspended sign imported from another convention.

Table 1 gives the carrier map needed to interpret the row indices. The retained gravity representatives are dressed combinations, so the table states their image by bundle type rather than pretending that the gravity part of $\iota$ is a coordinate inclusion. The Maxwell summand is retained by the identity.

Table 1: Human-readable carrier map. A zero retained image means that the displayed full block belongs to the contractible complement.

| Full rows | Degree | Full carrier content | Retained image |
|---|---|---|---|
| 0–4 | $-1$ | three spatial diffeomorphism ghosts, clock ghost $\tau$, Weyl ghost $\sigma$ | spatial ghosts 0–2 |
| 5–16 | 0 | ten $\widehat{h}_{\mu\nu}$ fields, $R$, $\Theta$ | dressed $\widehat{h}$ rows 3–12 |
| 17–26 | 0 | nonminimal multipliers and antighost-duals | 0 |
| 27–37 | 1 | ten $\widehat{h}^*_{\mu\nu}$ equations and $R^*$ | dressed $\widehat{h}^*$ rows 13–22 |
| **38** | 1 | $\boldsymbol{\Theta^*}$, the clock/Stueckelberg equation dual | **0** under $\pi$ |
| 39–48 | 1 | nonminimal dual and antighost rows | 0 |
| 49–53 | 2 | three spatial ghost antifields, $\tau^*$, $\sigma^*$ | spatial ghost-antifields 23–25 |
| 54 | $-1$ | Maxwell ghost $c_M$ | $c_M$ (row 26) |
| 55–58 | 0 | Maxwell fields $A_0, \dots, A_3$ | rows 27–30 |
| 59–62 | 1 | Maxwell equations $A_0^+, \dots, A_3^+$ | rows 31–34 |
| 63 | 2 | Maxwell ghost antifield $c_M^+$ | row 35 |

The row-38 support statement is therefore geometric. Rows $3, 16, 38, 52$ of the full carrier are $\tau, \Theta, \Theta^*, \tau^*$, and the certified blocks include

$$\mathrm{pr}_\Theta\, q_1(\tau) = \Theta, \qquad q_1(\Theta^*) = -\tau^*, \qquad S(\Theta) = \tau, \qquad S(\tau^*) = -\Theta^*. \tag{9}$$

Moreover

$$\pi\big|_{\langle \Theta^*, \tau^* \rangle} = 0, \qquad \mathrm{pr}_{\langle \Theta^*, \tau^* \rangle}\, \iota = 0, \qquad \omega_{64}(\Theta, \Theta^*) = 1, \quad \omega_{64}(\tau, \tau^*) = 1. \tag{10}$$

Thus row 38 is one half of the cyclic dual contractible clock doublet, not an unnamed discarded equation. At block level, $\iota$, $\pi$, and $S$ have respectively 86, 86, and 19 nonzero row-pair blocks; the vanishing in (10) is checked against those complete support ledgers.

The based classical BV vector field is known through cubic Taylor order,

$$q = q_1 + q_2 + q_3 + O(4), \tag{11}$$

in the repository's suspended graded-symmetric factorial convention. In this convention every $q_n$ and $\ell_n$ has cohomological degree $+1$. The full tensors are action-derived, support-local, typed cyclic, and equivariant under the background-fixing helical generator

$$K_{\text{Berger}} = D - \omega R. \tag{12}$$

Raw cylinder translation $D$ is affine at the rotating background and is not substituted for $K_{\text{Berger}}$.

The result combines two dependency layers:

| Layer | Certified content | Explicit limitation |
|---|---|---|
| LOCAL-ALGEBRAIC | exact $q_2, q_3, \ell_2, \ell_3$, identities and cyclicity | no residual branch interpretation |
| LORENTZIAN-CAUSAL | unary Green homotopies and $K$-Cartan support through arity three | no Hadamard products or QME |

Neither layer is silently promoted to RESIDUAL-TRANSFERRED or LORENTZIAN-CERTIFIED quantum status.

# 3 Ternary homological transfer

**Proposition 3.1** (Formal cyclic transfer through arity three)**.** *Let a cyclic $L_\infty$ algebra be given through $q_3$, and let $(\pi, \iota, S)$ be a cyclic SDR satisfying (5)–(6) and (7). The rooted-tree formulas (14)–(17) define cyclic operations through ternary order and imply*

$$[\ell_1, \ell_3] + \frac{1}{2}[\ell_2, \ell_2] = 0. \tag{13}$$

This is the standard cyclic homotopy-transfer theorem [3, 4]. Its application is the mathematical argument for the identities. The sparse calculations below are a separate implementation certificate: they test that the declared signs, PBW normalization, carrier maps, and typed pairing actually realize the formal theorem.

The first two transferred Taylor operations are

$$\ell_1 = \pi q_1 \iota, \tag{14}$$

$$\ell_2 = \pi q_2(\iota, \iota). \tag{15}$$

Define the second Taylor component of the inclusion by

$$I_2 = -S q_2(\iota, \iota). \tag{16}$$

Then the ternary transfer formula is

$$\ell_3 = \pi q_3(\iota, \iota, \iota) + \sum_{\sigma \in \text{Sh}(2,1)} \epsilon(\sigma)\, \pi q_2\big(I_2(\iota_{\sigma(1)}, \iota_{\sigma(2)}), \iota_{\sigma(3)}\big). \tag{17}$$

The first term is the contact contribution. The second line is the exchange contribution $q_2 S q_2$.

**Lemma 3.2** (Exact mixed exchange support for the frozen SDR)**.** *For the mixed gravity–Maxwell ternary operation, the gravity-outer/mixed-inner exchange channel contains* 144 *outer/inner coefficient pairs. The three graded unshuffles produce* 324 *signed contributions and* 342 *canonical full-complex PBW coefficients. Every one has full output row* $38 = \Theta^*$*. The retained projection has no support on that row, so this channel vanishes after projection. The mixed-outer/gravity-inner and mixed-outer/mixed-inner channels have no outer/inner pairs.*

*Proof.* The exact consumer reconstructs the 96 gravity and 12 mixed coefficients of $I_2$, including their full-row supports, before composing them with $q_2$. Outer Leibniz expansion can split one signed unshuffle contribution among several derivative distributions before canonical collection; this is why 324 signed contributions yield 342 PBW coefficients. Direct application of $\pi$ annihilates the 342 row-38 coefficients. As a negative control, adding the single mutant entry

$$\pi_{\mathrm{mut}}(\text{retained row } 0, \text{full row } 38) = 1 \tag{18}$$

exposes all 342 coefficients. Thus the zero is caused by the certified projection and is not an empty table hard-coded by the consumer. □

**Corollary 3.3** (Contact form for the frozen retained coordinates)**.** *The retained mixed exchange contribution is zero and*

$$\ell_3^{\mathrm{mixed}} = \pi q_3^{\mathrm{mixed}}(\iota, \iota, \iota). \tag{19}$$

*This equality is a computed property of the declared contraction, not a general truncation of the homotopy-transfer formula. In particular, a different cyclic homotopy or cyclic* $L_\infty$ *coordinate change may redistribute a representative between contact and exchange trees.*

# 4 Frozen-SDR nonvanishing and its audit

For a retained basis vector $u$, write $[\ell_3(x, y, z)]_u^{(0)}$ for the exact coefficient of $u$ with the zero PBW derivative word in every input slot.

**Theorem 4.1** (Nonzero retained mixed bracket for the frozen cyclic SDR)**.** *On the frozen positive Berger-clock gravity–Maxwell BV complex, the typed cyclic contraction* (5) *transfers the action-derived Taylor data through ternary order. Its mixed ternary representative is nonzero, with the two exact evaluations*

$$[\ell_3(\widehat{h}_{13}, A_1, A_3)]_{\widehat{h}_{00}^*}^{(0)} = +1, \tag{20}$$

$$[\ell_3(\widehat{h}_{01}, \widehat{h}_{03}, A_3)]_{A_1^+}^{(0)} = -1. \tag{21}$$

*The first is a gravity-equation output with one gravity and two Maxwell inputs; the second is a Maxwell-equation output with two gravity and one Maxwell input. Pairing the outputs with* $\widehat{h}_{00}$ *and* $A_1$*, respectively, makes both witnesses explicit coefficients in the lowered* $g, g, A, A$ *quartic sector. All mixed gravity–Maxwell exchange channels contributing to this declared* $\ell_3^{\mathrm{mixed}}$ *vanish for the frozen SDR by Lemma 3.2. The operations obey the arity-three* $L_\infty$ *identity* (13) *and cyclicity under the retained pairing.*

*Proof.* The two displayed coefficients are read from the exact independently replayed retained tensor; their output rows are 13 and 32, and their input rows are $(9, 28, 30)$ and $(4, 6, 30)$. All inputs have degree zero, both outputs have degree one, and the pairing partners are retained rows 3 and 28.

Hence the evaluations are typed and nonzero over $\mathbb{Q}(\sqrt{10})$ without reference to a total coefficient count. Lemma 3.2 proves the scoped exchange statement. Proposition 3.1 and the certified cyclic SDR give the abstract $L_\infty$ and cyclic conclusions. $\square$

The gravity labels in Equations (20) and (21) denote dressed retained coordinates. Their exact full-carrier inclusion columns, including every nonminimal gauge-fixing tail, are printed in Table 2. The Maxwell columns are identity columns. This prevents the witness notation from being mistaken for a raw-coordinate inclusion.

**Proposition 4.2** (Constant-field cyclic-redefinition screen)**.** *Let* $\mathrm{ev}_0$ *set every PBW derivative word to zero after lowering the retained operations with the typed odd pairing. On the complete mixed quartic space*

$$\mathrm{Sym}^2(G^*) \otimes \mathrm{Sym}^2(A^*), \qquad \dim = 550, \tag{22}$$

*the matter-parity-preserving, zero-jet BV-canonical cotangent-lift ansatz has* 810 *coefficients in* $F_2$ *and* 1,880 *in* $F_3$*. Its infinitesimal coboundary map*

$$\delta S_4 = \sum_i (\partial_i S_2) F_3^i + \sum_i (\partial_i S_3) F_2^i \tag{23}$$

*has exact rank* 550 *over* $\mathbb{Q}(\sqrt{10})$*. The landed constant-field mixed quartic has* 63 *nonzero coordinates and is reconstructed exactly by an exported* 51*-coefficient primitive supported entirely in* $F_3$*.*

*Proof.* The exact solver aggregates the frozen factorial-convention tensors into commutative lowered action polynomials, differentiates $S_2$ and $S_3$, and enumerates all 2,690 matter-parity-preserving base-field maps. The resulting $550 \times 2{,}690$ matrix has 5,105 nonzero entries. A deterministic support matching selects a 550-column block whose rank is then checked exactly; no structural or floating-point rank is used. The independent verifier rebuilds the action polynomials and coboundary matrix from the pinned $\ell_1, \ell_2, \ell_3$ payloads, rechecks the rank basis, and multiplies the exported primitive into all 550 target coordinates. A one-coefficient primitive mutation fails the reconstruction. $\square$

Proposition 4.2 is a G0 filtration result, not a trivialization of the full retained operation. In particular, derivative terms in $F_2, F_3$, PBW integration-by-parts relations, and cancellation of higher-order terms against the fourth-order unary operator have not yet been solved, and the 288 ghost/antifield completion coefficients have not been independently matched by this primitive.

**Proposition 4.3** (First positive-jet cyclic-redefinition screen)**.** *After exact first-order integration by parts, the mixed physical quartic has* 1,330 *first-jet coordinates per PBW axis. Let* $A$ *denote the zero-jet map of Proposition 4.2,* $B_a$ *the first-jet effect of its complete* 2,690*-coefficient zero-jet ansatz, and* $C$ *the map from first-jet redefinitions. The coupled equations*

$$Ax_0 = t_0, \qquad B_a x_0 + C y_a = t_{1,a}, \quad a = 0, 1, 2, 3, \tag{24}$$

*have an exact solution over* $\mathbb{Q}(\sqrt{10})$*. More precisely,* $C$ *has shape* $1{,}330 \times 6{,}560$*, rank* 1,327*, and cokernel dimension three. The cokernel-reduced Schur system has shape* $562 \times 2{,}690$*, rank* 557*, and compatible target. An exported primitive has* 51 *zero-jet coefficients and respectively* 43, 94, 95, 108 *first-jet coefficients on the four PBW axes.*

*Proof.* The solver lowers the pinned $\ell_1, \ell_2, \ell_3$ tensors to local action densities and quotients total first derivatives with exact field-multiplicity weights. It projects Equation (24) to the three exact cokernel rows of $C$ on each axis before solving for $x_0$; hence it retains the entire affine zero-jet solution space rather than extending the earlier chosen primitive by fiat. Exact rank receipts and direct multiplication of the exported $x_0, y_a$ reconstruct the 550 zero-jet coordinates and all four 1,330-coordinate first-jet blocks. A primitive-coefficient mutation fails the replay. $\square$

This remains a physical-action filtration theorem. It does not match the 288 ghost/antifield completion coefficients and does not address total jet order two, where the landed $\ell_3$ has nonzero terms.

**Corollary 4.4** (Exact sparse implementation audit)**.** *The independent PBW consumer realizes Theorem 4.1 with the following basis-level audit data.*

1. *The retained coupled binary operation $\ell_2$ contains* 1,474 *canonical coefficients.*
2. *The retained mixed ternary operation $\ell_3$ contains* 25,950 *canonical coefficients on* 18 *nonzero output rows.*
3. *All mixed gravity–Maxwell exchange channels contributing to the declared mixed $\ell_3$ vanish for the support reason in Lemma 3.2.*
4. *Direct coefficient comparison of the retained arity-three identity closes on all* 36 *rows:*

$$[\ell_1, \ell_3] + \frac{1}{2}[\ell_2, \ell_2] = 0. \tag{25}$$

5. *The mixed bracket has* 7,614 *gravity-output coefficients with two Maxwell inputs and* 18,336 *Maxwell-output coefficients with one Maxwell input.*

*A localized one-coefficient mutation of $\ell_3$ creates two exact arity-three defects.*

The audit is obtained as follows. Equations (15)–(17) are evaluated in exact sparse PBW normal form over $\mathbb{Q}(\sqrt{10})$. The independent consumer parses all 59,598 coefficients of the full mixed $q_3$ payload and reproduces all 25,950 retained contact coefficients with no missing, extra, or changed terms. Lemma 3.2 evaluates every possible mixed exchange channel. Direct comparison of (25) gives zero defects on all rows. The input-sector and output-sector ledgers give the two sector counts. Finally, the mutation changes one retained coefficient and produces two normalized defects, preventing acceptance by a consumer that merely assumes the expected answer.

The numbers in Corollary 4.4 are reproducibility metadata, not invariant content: they depend on the retained basis, row ordering, PBW and integration-by-parts normalization, and the chosen SDR. The invariantly stronger questions in Equation (2) remain open. In particular, Theorem 4.1 is weaker than a residual particle vertex because $\mathcal{R}_{36}$ is retained, not minimal or decomposed into its dynamical branches.

# 5 Independent cyclic implementation audits

Lowering the output of $\ell_3$ with the odd pairing gives a four-linear functional. For degree-zero retained inputs, cyclic transposition of the first input requires formal adjunction of every PBW derivative through the invariant volume form. If $w$ is a derivative word, the exact formal adjoint distributes its letters over the other three slots with sign $(-1)^{|w|}$ and the corresponding multinomial multiplicities.

**Proposition 5.1** (Independent degree-zero lowered cyclicity)**.** *Every degree-zero coefficient of the lowered retained $\ell_3$ satisfies exact cyclic transposition. There are*

$$25{,}662 = 7{,}506_{\text{gravity output}} + 18{,}156_{\text{Maxwell output}} \tag{26}$$

*such coefficients, and the defect count is zero on every row. Replacing the Maxwell field–equation weight two by weight one produces* 17,108 *defects on* 14 *rows.*

*Proof.* The verifier constructs both sides independently from the retained tensor, the typed pairing, and exact PBW integration by parts. The signed pairing orientations are retained as a separate convention ledger; only their absolute component multiplicities enter the degree-zero field–equation transpose. Equality holds coefficientwise. The weight mutation changes no coefficient of $\ell_3$ itself and therefore tests the lowering convention rather than the imported interaction. □

The remaining

$$25{,}950 - 25{,}662 = 288 \tag{27}$$

coefficients are the ghost/antifield completion. Their retained output rows are

$$23,\ 24,\ 25,\ 26,\ 32,\ 33,\ 34.$$

If $x$ is the transposed first input and $u$ is the coordinate paired with the output, the suspended-Darboux sign is

$$(-1)^{|x||u|+\epsilon_2(x)+\epsilon_2(u)}, \qquad \epsilon_2(v) = \begin{cases} 1, & \deg v = 2, \\ 0, & \text{otherwise.} \end{cases} \tag{28}$$

Here the degree-two rows are precisely the retained ghost-antifield coordinates.

**Proposition 5.2** (Independent full-BV quartic cyclicity)**.** *The complete* 25,950*-coefficient lowered retained* $\ell_3$ *satisfies exact cyclic transposition. Within the* 288*-coefficient completion,* 120 *coefficients carry the positive transpose sign and* 168 *carry the negative sign. The exact defect count is zero. Omitting the degree-two polarization term exposes* 132 *defects on all seven completion output rows.*

This proposition is a full-BV statement about the pinned classical retained tensor. It does not imply full BV local cohomology, a quantum master equation, or a Lorentzian quantum theory.

# 6 Causal compatibility through ternary order

The causal statement is not inferred from finite PBW tables. It is supplied by the independently certified unary Green chain contractions and the coupled cyclic Cartan theorem, extending the Berger-clock construction of the companion paper [8]. If

$$\Lambda_{36,\pm} = \Lambda_{26,\pm}^{\text{grav+clock}} \oplus \Lambda_{M,\pm}, \qquad \Lambda_{M,\pm} = W_M G_{P_M,\pm}, \tag{29}$$

then the certified complete-carrier lift is

$$\Lambda_{64,\pm} = S + \iota\Lambda_{36,\pm}\pi, \qquad q_1\Lambda_{64,\pm} + \Lambda_{64,\pm}q_1 = \mathrm{id}_{64}. \tag{30}$$

The side conditions imply the explicit retained bridge

$$\pi\Lambda_{64,\pm}\iota = \Lambda_{36,\pm}. \tag{31}$$

The advanced and retarded homotopies have the corresponding one-sided causal supports and formal-adjoint reversal. Equivariance is not inferred from the background label: the exact carrier ledgers verify

$$K_{64}\iota = \iota K_{36}, \qquad \pi K_{64} = K_{36}\pi, \qquad K_{64}S = SK_{64}. \tag{32}$$

Let $\iota_K^{(1)}$ be the unary Cartan primitive. The higher sources are

$$A_K^{(2)} = [q_2, \iota_K^{(1)}] - L_K^{(2)}, \tag{33}$$

$$A_K^{(3)} = [q_3, \iota_K^{(1)}] + [q_2, \iota_K^{(2)}] - L_K^{(3)}. \tag{34}$$

At the frozen background $L_K^{(2)} = L_K^{(3)} = 0$. The $L_\infty$ identities, graded Jacobi identity, and $K$-equivariance give

$$[q_1, A_K^{(2)}] = [q_1, A_K^{(3)}] = 0. \tag{35}$$

Raw causal primitives are cyclicized with the exact Reynolds projectors

$$\mathrm{Cyc}_3 = \frac{1+\tau+\tau^2}{3}, \qquad \mathrm{Cyc}_4 = \frac{1+\tau+\tau^2+\tau^3}{4}. \tag{36}$$

**Theorem 6.1** (Cyclic causal Cartan compatibility)**.** *The coupled 64-row complex and its typed 36-row retained contraction obey the $K_{\mathrm{Berger}}$ Cartan identities through arity three. The higher cyclic primitives satisfy*

$$\mathrm{supp}\big(\iota_K^{(n)}(f_1, \dots, f_n)\big) \subseteq J\left(\bigcup_i \mathrm{supp}\, f_i\right), \qquad n \leq 3. \tag{37}$$

*The Taylor operations are support-local and no spatial inverse or mode projector enters the construction.*

The higher cyclic primitives use the two-sided causal hull. They are not claimed to be separately retarded or advanced; the unary homotopies retain that stronger one-sided property. Equations (30)–(32) make the bridge between the complete and retained unary causal data explicit. They do not turn the support-local algebraic homotopy $S$ used in (17) into a causal propagator. Thus Theorem 6.1 is a compatibility result parallel to Theorem 4.1, not a construction of $\ell_3$ by causal integration. It does not construct interacting retarded products, a Hadamard state, or a Lorentzian quantum master equation.

# 7 Residual descent and the projector obstruction

The next calculation determines what the surviving tensor means physically. On any admissible enlarged or alternative carrier, the dynamical sector must contain

$$\mathcal{R}_{\mathrm{dyn}} = \mathcal{R}_{\mathrm{Einstein}} \oplus \mathcal{R}_{\mathrm{extra\ Weyl}} \oplus \mathcal{R}_{\mathrm{Maxwell}}, \tag{38}$$

together with exact inclusion, projection, pairing, parity, real structure, and $K_{\mathrm{Berger}}$ weights. The mixing entries are then

$$M^a{}_{bcd} = \pi^a_{\mathrm{dyn}}\, \ell_3\big(\iota_b^{\mathrm{dyn}}, \iota_c^{\mathrm{dyn}}, \iota_d^{\mathrm{dyn}}\big). \tag{39}$$

Every claimed zero must carry an exact support or cancellation witness, and graded symmetry and cyclicity must be replayed in branch coordinates.

The canonical attempt on $\mathcal{R}_{36}$ has received a binary negative verdict. With the Einstein image fixed by the certified rough tensor-wave branch, the exact endpoint

$$A_{10} = \Box_2^2 + V_2, \qquad \mathrm{ord}(V_2) \leq 2, \tag{40}$$

has 92 nondivisible degree-two remainder entries. A normalized first witness is

$$\frac{71p_1^2 + 71p_2^2 + 9p_3^2}{80} \quad \mathrm{mod} \quad (-p_0^2 + p_1^2 + p_2^2 + p_3^2), \tag{41}$$

and the normalized functional $(80/71)[p_1^2]$ evaluates it to one. Hence no canonical finite-order support-local same-bundle complementary Einstein-like/extra-Weyl projector of the declared type exists on $\mathcal{R}_{36}$.

This is a scoped obstruction, not a global no-go. Mixed-bundle mapping cylinders, filtered carriers, higher-rank local carriers, and explicitly REDUCED-MODE nonlocal splittings remain open. Exact symbol counting requires at least four additional BV rows; the smallest natural support-local candidate adds a spatial STF2 prolongation and its cyclic dual, giving rank 46.

That candidate has now been constructed. Let $T_{\mathrm{STF}}$ extract

$$h_{12}, \quad h_{13}, \quad h_{23}, \quad h_{11} - h_{22}, \quad h_{11} + h_{22} - 2h_{33}, \tag{42}$$

and set $F = T_{\mathrm{STF}}\Box_2$. Introducing $Y = Fh$ and its five cyclic-dual rows gives degree ranks $(4, 19, 19, 4)$. The metric Hessian block is

$$\begin{pmatrix} A_{10} + F^\dagger F & -F^\dagger \\ -F & I_5 \end{pmatrix}, \tag{43}$$

whose Schur complement is exactly $A_{10}$.

**Proposition 7.1** (Exact rank-46 STF2 graph carrier)**.** *The finite-order cotangent shear defining Equation (43) and its inverse produce a cyclic strong deformation retract*

$$(\mathcal{R}_{46}, q_1^{46}, \omega_{46}) \underset{\iota_{36}^{46}}{\overset{\pi_{46}^{36}}{\rightleftarrows}} (\mathcal{R}_{36}, \ell_1, \omega_{36}), \qquad S_{46}, \tag{44}$$

*with*

$$(q_1^{46})^2 = 0, \qquad \pi_{46}^{36}\iota_{36}^{46} = I_{36}, \qquad q_1^{46}S_{46} + S_{46}q_1^{46} = I_{46} - \iota_{36}^{46}\pi_{46}^{36},$$
$$S_{46}^2 = 0, \qquad S_{46}\iota_{36}^{46} = 0, \qquad \pi_{46}^{36}S_{46} = 0.$$

*Both chain maps, the induced pairing, unary cyclicity, and homotopy cyclicity hold exactly over $\mathbb{Q}(\sqrt{10})$. The added STF2 complement is contractible, so this construction changes neither the retained cohomology nor, by itself, the physical branch content.*

The graph extension does not, however, split the repeated physical wave layer at principal order. Put

$$A = \mathbb{Q}(\sqrt{10})[\epsilon]/(\epsilon^2), \qquad \epsilon = -p_0^2 + p_1^2 + p_2^2 + p_3^2. \tag{45}$$

For each physical helicity, the Einstein-like layer is the submodule $E = \epsilon A$ and the generalized extra-Weyl layer is the quotient $A/E$.

**Proposition 7.2** (Normalized principal branch-anchor obstruction)**.** *The exact sequence*

$$0 \longrightarrow \epsilon A \longrightarrow A \longrightarrow A/(\epsilon) \longrightarrow 0 \tag{46}$$

*does not split $A$-linearly. Indeed, an attempted section has $s(1) = 1 + b\epsilon$, whereas $A$-linearity requires $\epsilon s(1) = 0$ and exact multiplication gives*

$$\epsilon s(1) = \epsilon. \tag{47}$$

*The functional extracting the coefficient of $\epsilon$ evaluates this defect to one and annihilates every correction $b\epsilon$. Equivalently, an endomorphism $a + b\epsilon$ is idempotent only for $(a, b) = (0, 0)$ or $(1, 0)$. The real helicity multiplicity is two and the witness has $K_{\mathrm{Berger}}$-weight zero.*

This obstruction excludes a nonzero principal direct-sum anchor and prevents the zero, identity, or contractible-auxiliary projector from being promoted as the physical branch split. It is not a no-go for the full rank-46 projector: lower-order Berger terms may deform the repeated leading factor and split it.

The physical quotient needed for that next calculation is now exact. On the normalized projective null cone, the tensorial transverse–traceless idempotent defines a rank-two projective polarization module inside the full six-dimensional degree-zero Berger null-symbol cohomology. Its canonical transpose defines the paired degree-one module, and the odd BV pairing descends nondegenerately. At $\zeta = (1, 1, 0, 0)$ its normalized basis is $\widehat{h}_{22} - \widehat{h}_{33}, \widehat{h}_{23}$ and agrees with the independently certified linearized-Weyl quotient. Thus the polarization rank is two, while tensoring with $\mathbb{Q}(\sqrt{10})[\epsilon]/(\epsilon^2)$ gives a rank-four generalized wave module. No global two-column helicity frame is chosen: the invariant object is the projective module defined by the idempotent.

The filtered order-two extension class defined by $V_2$ has now been tested by the full symbol-complex descent equation. At $\zeta = (1, 1, 0, 0)$, arbitrary principal gauge changes of the field representative, all principal Hessian boundaries, and both physical equation representatives are included. A normalized left-null witness evaluates on the two physical columns as

$$(1, 0)\,. \tag{48}$$

Thus the plus representative $\widehat{h}_{22} - \widehat{h}_{33}$ has a nonzero quotient class for every allowed correction. The cross representative $\widehat{h}_{23}$ does lift, with physical equation coefficient 71/40. This is an invariant filtered obstruction, not a diagonalization of the raw $10 \times 10$ matrix or the bare compression $\Pi_{\mathrm{TT}} V_2 \Pi_{\mathrm{TT}}$.

The declared rank-46 contractible graph carrier therefore does not supply a closed physical filtered subcomplex or an Einstein-like/extra-Weyl projector with this principal anchor. It also does not supply a $q_2, q_3$ lift, $K_{\mathrm{Berger}}$ equivariance, or causal support on $\mathcal{R}_{46}$. Equation (39) therefore remains a target formula rather than an authorized rank-46 calculation.

The receiving contract also reserves a separate deformation/vertex payload. That deformation complex, its normalized basis, and its Euler–Lagrange map are neither defined nor used in this paper. Their rank and topological statements are therefore deferred to the residual-mixing paper rather than introduced as a second theorem here. In particular, no deformation class is relabelled as a third dynamical particle branch.

The obstruction neither invalidates Theorem 4.1 nor computes a branch mixing table. It is scoped to the declared physical principal anchor on the exact contractible rank-46 graph carrier. It does not rule out a different leading cohomology carrier, a noncontractible or mixed-bundle curvature mapping cylinder, or a separately tagged nonlocal reduced-mode splitting. The next exact choice is therefore to retain the unsplit complex or construct the smallest noncontractible filtered enlargement before lifting the nonlinear brackets.

# 8 Filtered cyclic nonremovability

The constant-field and first-jet physical-action primitives do not decide the full BV deformation problem: cyclic completion couples fields, ghosts and antifields, and a choice of lower primitive may hide kernel directions needed on the next filtration page. We therefore use the summed pre-reduction input PBW order and solve the zero and first full-BV pages simultaneously. The allowed coordinate changes are degree-zero, derivative-aware cyclic super-cotangent Taylor maps $(F_2, F_3)$ of nonnegative filtered order.

Let $A_0$ denote the complete zero-page coboundary map, let $B_a$ be the order-one contribution of a zero-page map on derivative axis $a$, and let $C_a$ be the associated-graded first-jet coboundary. If $r_a$ is the first-page residual after one certified zero-page primitive, changing that primitive by $x$ and adding first-jet maps $y_a$ requires

$$A_0 x = 0, \qquad B_a x + C_a y_a = r_a, \qquad a = 0, 1, 2, 3. \tag{49}$$

The first equation retains the complete kernel of $A_0$; freezing the earlier 67-term primitive would give a false single-coordinate obstruction.

**Theorem 8.1** (Scoped filtered cyclic obstruction)**.** *For the frozen retained cyclic SDR, Equation* (49) *has no solution over* $\mathbb{Q}(\sqrt{10})$*. There is an exact normalized functional* $\mathfrak{w}$ *with* 22 *supported full-BV coefficient rows such that*

$$\mathfrak{w}(A_0) = 0, \qquad \mathfrak{w}(B_a, C_a) = 0 \quad (a = 0, 1, 2, 3), \qquad \mathfrak{w}(r) = 1. \tag{50}$$

*The exhaustive transpose replay checks all* 5,984 *zero-page base-label columns and all* 14,998 *first-jet columns on each axis with zero defects. Consequently no admissible nonnegative filtered cyclic* $F_2/F_3$ *transformation removes the mixed* $\ell_3$ *through summed PBW order two.*

The obstruction lives on the first associated-graded page. Higher filtered profiles have order at least two and cannot alter it, so extending the ansatz to $q_4$ is irrelevant to this verdict. The displayed 22-row functional and its coefficients depend on the chosen SDR coordinates and cokernel basis; the invariant content is the nonzero class in this declared filtered cyclic quotient. The theorem does not identify a class in an unrestricted cyclic Chevalley–Eilenberg complex, prove invariance under changing SDR, or descend the operation to $H(\mathcal{R}_{36}, \ell_1)$.

# 9 Interpretation and nonclaims

At representative level, the retained classical tensor contains both a gravity-equation response to one gravity and two Maxwell inputs and a Maxwell-equation response to two gravity and one Maxwell input. After lowering, both are $g, g, A, A$ quartic coefficients. They remain visible under the frozen exact cyclic transfer and satisfy the ternary homotopy identity. If a noncontractible enlargement receives an admissible branch projector and nonlinear lift—or an explicitly declared unsplit observable replaces the branch table— Equation (39) will decide whether this representative closes on the Einstein-like branch, couples it to the extra-Weyl branch, or vanishes on residual cohomology.

Several stronger conclusions do not follow.

1. The bracket changes no unary kinetic operator. It therefore introduces no new negative kinetic direction at this order, but this is not a unitarity or positive-Hilbert-space theorem.

2. A nonzero retained bracket is not yet a photon or graviton scattering amplitude. Residual projection, causal state selection, and asymptotic or relational observables are still required.

3. Nonvanishing of this coefficient representative alone proves neither a nonzero operation on $\ell_1$-cohomology nor a class in the unrestricted cyclic deformation complex. The complete constant-field page is removable by the exact $F_3$ primitive of Proposition 4.2, and the first-jet physical page is removable by Proposition 4.3. Theorem 8.1 nevertheless proves full-BV nonremovability in its declared filtered cyclic class. No invariance under changing SDR is claimed.

4. Deformation/vertex classes are not one-particle states and are not included in the dynamical branch list used here.

5. The vanishing one-particle cohomology of the separate free conformal cylinder problem is not converted here into an interacting vacuum theorem.

6. No loop coefficient, anomaly cancellation, QME restoration, Hadamard state, renormalized product, or quantum interaction has been constructed.

7. No arity-four or convergent all-orders $L_\infty$ or Cartan theorem is claimed.

The appropriate provisional interpretation is therefore a retained deformation/vertex representative rather than a particle interaction. The mixing table and deformation-cohomology calculation are the two distinct next tests.

## 10 Reproducibility and certificate map

The algebraic theorem is pinned to the classical hardening commit `e99d0c1d39490de5261fc6ca1dc2aeaa0d149655`. Its principal artifacts are

| Artifact | Role |
|---|---|
| `BERGER_COUPLED_K_CARTAN_THROUGH_ARITY_THREE` | full/retained cyclic causal $K$-Cartan theorem |
| `BERGER_RETAINED_MIXED_ELL3_INDEPENDENT_ACCEPTANCE` | exact independent $64 \to 36$ transfer and mutation witnesses |
| `BERGER_RETAINED_MIXED_ELL3_PHYSICAL_CYCLICITY` | independent degree-zero lowered cyclicity and pairing-weight mutation |
| `BERGER_RETAINED_MIXED_ELL3_FULL_BV_CYCLICITY` | independent ghost/antifield completion and full retained BV cyclicity |
| `BERGER_RETAINED_MIXED_ELL3_CONSTANT_FIELD_REDEFINITION_V1` | exact constant-field physical-action cyclic-redefinition primitive |
| `BERGER_RETAINED_MIXED_ELL3_FIRST_JET_REDEFINITION_V1` | exact coupled first-jet physical-action primitive |
| `BERGER_RETAINED_MIXED_ELL3_POSITIVE_JET_FULL_BV_OBSTRUCTION_V1` | normalized 22-row obstruction to the declared filtered cyclic removal |
| `NONLINEAR_SOURCE_TRANSFER_TANGENT_CONE_DICTIONARY_V1` | $D^2E$–$q_2$–$\ell_2$–$\ell_3$ normalization and obstruction naturality |
| `BERGER_RESIDUAL_MIXED_ELL3_BRANCH_PROJECTION_READINESS_V2` | historical fail-closed input contract for the mixing table |
| `BERGER_RETAINED_36_RESIDUAL_BRANCH_LOCAL_PROJECTOR_OBSTRUCTION_V1` | normalized obstruction to the canonical local 36-row projector |
| `BERGER_RETAINED_46_STF2_PROLONGATION_BRANCH_CARRIER_V1` | exact local cyclic graph carrier; physical-anchor projector tested below |
| `BERGER_RETAINED_46_STF2_PRINCIPAL_BRANCH_ANCHOR_V1` | normalized principal nonsplitting witness |
| `BERGER_RETAINED_46_STF2_PHYSICAL_HELICITY_FILTERED_QUOTIENT_V1` | exact rank-two projective helicity module inside the six-class null quotient |
| `BERGER_RETAINED_46_STF2_SUBPRINCIPAL_BRANCH_ANCHOR_OR_OBSTRUCTION_V1` | normalized $V_2$ filtered-lift obstruction for the plus polarization |

The scoped reproduction commands are

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_coupled_k_cartan_through_arity_three.py --check --guards
PYTHONPATH=. python3 \
```

```
  d_quotient_classical/backreacted_clock/\
  verify_berger_coupled_k_cartan_through_arity_three.py
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_acceptance_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_acceptance
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_physical_cyclicity_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_physical_cyclicity
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_ghost_antifield_cyclicity_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_ghost_antifield_cyclicity
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_mixed_ell3_constant_field_redefinition.py --check
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_mixed_ell3_constant_field_redefinition.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_mixed_ell3_first_jet_redefinition.py --check
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_mixed_ell3_first_jet_redefinition.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_mixed_ell3_positive_jet_full_bv_obstruction.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_mixed_ell3_positive_jet_full_bv_obstruction_exhaustive.py \
  --workers 8
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_nonlinear_source_transfer_tangent_cone_dictionary.py
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_residual_ell3_branch_projection_readiness_v2 --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_residual_ell3_branch_projection_readiness_v2
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_36_residual_branch_local_projector_obstruction.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_46_stf2_prolongation_branch_carrier.py --check --guards
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_46_stf2_prolongation_branch_carrier.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_46_stf2_principal_branch_anchor.py --check --guards
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_46_stf2_principal_branch_anchor.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_46_stf2_physical_helicity_filtered_quotient.py \
  --check --guards
```

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_46_stf2_physical_helicity_filtered_quotient.py
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_retained_46_stf2_subprincipal_branch_anchor_or_obstruction.py \
  --check --guards
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_retained_46_stf2_subprincipal_branch_anchor_or_obstruction.py
```

The companion machine-readable claim map records the dependency hashes, positive claims, and explicit nonclaims. The scoped frozen-SDR transfer and filtered-cyclic obstruction statements are theorem-frozen; specialist prose and artifact review remain before release.

## 11 Outlook

Paper 11 now has two mathematical continuations. The first is to decide between the unsplit retained complex and a noncontractible filtered or mixed-bundle enlargement, then materialize the relevant cyclic $q_2, q_3$ lift and compute the corresponding version of Equation (39). A nonzero Einstein/extra-Weyl entry would turn the additional Weyl physics into an explicit interaction channel; a block-diagonal table would establish nonlinear closure of the Einstein-like direction through this order; and a central topological action would isolate the vertex theory from the dynamical carrier. The second is to test whether the selected mixed representative defines a nonzero operation on $\ell_1$-cohomology or a class in an unrestricted cyclic deformation complex. Its constant-field and first-jet physical-action pages are exact, while the complete full-BV filtered screen is obstructed on its first associated-graded page by Theorem 8.1. The next deformation calculation is therefore residual descent or comparison with a larger admissible equivalence class, not an enlargement to $q_4$.

The algebraic theorem does not need to wait for that verdict. What the mixing table changes is its physical interpretation, not the existence of the nonzero mixed representative for the frozen cyclic SDR.

## A Exact inclusion columns for the displayed witnesses

In Table 2, $\alpha = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ denotes the ordered PBW derivative word in the four-generator order of the carrier certificate. The table is generated deterministically from the complete $64 \times 36$ inclusion record; it contains 106 exact operator coefficients. The five columns shown are the three dressed gravity inputs used by Equations (20)–(21), the gravity-equation output, and its dressed pairing partner.

Table 2: Exact full-carrier inclusion columns for every dressed gravity coordinate used by the two headline witnesses or their pairing partners. Each row denotes the term $c\,\partial^\alpha e_{\mathrm{full}}$ in $\iota(e_{\mathrm{retained}})$.

| Retained ID | Row | Full-carrier ID | $\alpha$ | Exact $c$ |
|---|---|---|---|---|
| h_hat_13 | 9 | h_hat_13 | $(0,0,0,0)$ | 1 |
| | | bar_c_star_diff_1 | $(2,0,0,1)$ | -1 |
| | | bar_c_star_diff_1 | $(0,2,0,1)$ | 1/3 |
| | | bar_c_star_diff_1 | $(0,1,1,0)$ | -11*sqrt(10)/45 |
| | | bar_c_star_diff_1 | $(0,0,2,1)$ | 1 |
| | | bar_c_star_diff_1 | $(0,0,0,1)$ | -475/48 |

Continued on next page

Table 2 (continued)

| Retained ID | Row | Full-carrier ID | $\alpha$ | Exact $c$ |
|---|---|---|---|---|
| | | bar_c_star_diff_1 | $(0,0,0,3)$ | 1 |
| | | bar_c_star_diff_2 | $(0,0,0,0)$ | -79391*sqrt(10)/43200 |
| | | bar_c_star_diff_2 | $(2,0,0,0)$ | -31*sqrt(10)/60 |
| | | bar_c_star_diff_2 | $(0,2,0,0)$ | 11*sqrt(10)/30 |
| | | bar_c_star_diff_2 | $(0,1,1,1)$ | -2/3 |
| | | bar_c_star_diff_2 | $(0,0,2,0)$ | 11*sqrt(10)/90 |
| | | bar_c_star_diff_2 | $(0,0,0,2)$ | 9*sqrt(10)/5 |
| | | bar_c_star_diff_3 | $(2,1,0,0)$ | -1 |
| | | bar_c_star_diff_3 | $(0,1,0,0)$ | -329/2160 |
| | | bar_c_star_diff_3 | $(0,3,0,0)$ | 1 |
| | | bar_c_star_diff_3 | $(0,1,2,0)$ | 1 |
| | | bar_c_star_diff_3 | $(0,1,0,2)$ | 1/3 |
| | | bar_c_star_diff_3 | $(0,0,1,1)$ | 2*sqrt(10)/45 |
| | | bar_c_star_tau | $(1,1,0,1)$ | -2/3 |
| | | bar_c_star_tau | $(1,0,1,0)$ | -71*sqrt(10)/180 |
| | | bar_c_star_sigma | $(2,1,0,1)$ | 1/3 |
| | | bar_c_star_sigma | $(2,0,1,0)$ | 71*sqrt(10)/360 |
| | | bar_c_star_sigma | $(0,3,0,1)$ | -1/3 |
| | | bar_c_star_sigma | $(0,2,1,0)$ | -71*sqrt(10)/360 |
| | | bar_c_star_sigma | $(0,1,2,1)$ | -1/3 |
| | | bar_c_star_sigma | $(0,1,0,1)$ | 34/9 |
| | | bar_c_star_sigma | $(0,1,0,3)$ | -1/3 |
| | | bar_c_star_sigma | $(0,0,1,0)$ | 71*sqrt(10)/81 |
| | | bar_c_star_sigma | $(0,0,3,0)$ | -71*sqrt(10)/360 |
| | | bar_c_star_sigma | $(0,0,1,2)$ | -13*sqrt(10)/24 |
| h_hat_01 | 4 | h_hat_01 | $(0,0,0,0)$ | 1 |
| | | bar_c_star_diff_1 | $(1,0,0,0)$ | 2561/720 |
| | | bar_c_star_diff_1 | $(3,0,0,0)$ | 1 |
| | | bar_c_star_diff_1 | $(1,2,0,0)$ | -1/3 |
| | | bar_c_star_diff_1 | $(1,0,2,0)$ | -1 |
| | | bar_c_star_diff_1 | $(1,0,0,2)$ | -1 |
| | | bar_c_star_diff_2 | $(1,1,1,0)$ | 2/3 |
| | | bar_c_star_diff_2 | $(1,0,0,1)$ | -77*sqrt(10)/60 |
| | | bar_c_star_diff_3 | $(1,1,0,1)$ | 2/3 |
| | | bar_c_star_diff_3 | $(1,0,1,0)$ | 107*sqrt(10)/180 |
| | | bar_c_star_tau | $(2,1,0,0)$ | -1/3 |
| | | bar_c_star_tau | $(0,1,0,0)$ | -31/9 |
| | | bar_c_star_tau | $(0,3,0,0)$ | 1 |
| | | bar_c_star_tau | $(0,1,2,0)$ | 1 |
| | | bar_c_star_tau | $(0,1,0,2)$ | 1 |
| | | bar_c_star_tau | $(0,0,1,1)$ | 31*sqrt(10)/30 |
| | | bar_c_star_sigma | $(3,1,0,0)$ | -1/3 |
| | | bar_c_star_sigma | $(1,1,0,0)$ | -31/27 |
| | | bar_c_star_sigma | $(1,3,0,0)$ | 1/3 |
| | | bar_c_star_sigma | $(1,1,2,0)$ | 1/3 |
| | | bar_c_star_sigma | $(1,1,0,2)$ | 1/3 |
| | | bar_c_star_sigma | $(1,0,1,1)$ | 31*sqrt(10)/90 |
| h_hat_03 | 6 | h_hat_03 | $(0,0,0,0)$ | 1 |
| | | bar_c_star_diff_1 | $(1,1,0,1)$ | 2/3 |
| | | bar_c_star_diff_1 | $(1,0,1,0)$ | -3*sqrt(10)/20 |
| | | bar_c_star_diff_2 | $(1,1,0,0)$ | 3*sqrt(10)/20 |
| | | bar_c_star_diff_2 | $(1,0,1,1)$ | 2/3 |
| | | bar_c_star_diff_3 | $(1,0,0,0)$ | 9/80 |
| | | bar_c_star_diff_3 | $(3,0,0,0)$ | 1 |
| | | bar_c_star_diff_3 | $(1,2,0,0)$ | -1 |
| | | bar_c_star_diff_3 | $(1,0,2,0)$ | -1 |
| | | bar_c_star_diff_3 | $(1,0,0,2)$ | -1/3 |
| | | bar_c_star_tau | $(2,0,0,1)$ | -1/3 |
| | | bar_c_star_tau | $(0,2,0,1)$ | 1 |
| | | bar_c_star_tau | $(0,0,2,1)$ | 1 |
| | | bar_c_star_tau | $(0,0,0,3)$ | 1 |
| | | bar_c_star_sigma | $(3,0,0,1)$ | -1/3 |
| | | bar_c_star_sigma | $(1,2,0,1)$ | 1/3 |

Continued on next page

Table 2 (continued)

| Retained ID | Row | Full-carrier ID | $\alpha$ | Exact $c$ |
|---|---|---|---|---|
| | | bar_c_star_sigma | $(1,0,2,1)$ | 1/3 |
| | | bar_c_star_sigma | $(1,0,0,3)$ | 1/3 |
| h_hat_star_00 | 13 | h_hat_star_00 | $(0,0,0,0)$ | 1 |
| h_hat_00 | 3 | h_hat_00 | $(0,0,0,0)$ | 1 |
| | | bar_c_star_diff_1 | $(2,1,0,0)$ | -1/2 |
| | | bar_c_star_diff_1 | $(0,1,0,0)$ | 71/480 |
| | | bar_c_star_diff_1 | $(0,3,0,0)$ | 1/6 |
| | | bar_c_star_diff_1 | $(0,1,2,0)$ | 1/6 |
| | | bar_c_star_diff_1 | $(0,1,0,2)$ | 1/6 |
| | | bar_c_star_diff_2 | $(2,0,1,0)$ | -1/2 |
| | | bar_c_star_diff_2 | $(0,2,1,0)$ | 1/6 |
| | | bar_c_star_diff_2 | $(0,1,0,1)$ | -sqrt(10)/20 |
| | | bar_c_star_diff_2 | $(0,0,1,0)$ | -3/160 |
| | | bar_c_star_diff_2 | $(0,0,3,0)$ | 1/6 |
| | | bar_c_star_diff_2 | $(0,0,1,2)$ | 1/6 |
| | | bar_c_star_diff_3 | $(2,0,0,1)$ | -1/2 |
| | | bar_c_star_diff_3 | $(0,2,0,1)$ | 1/6 |
| | | bar_c_star_diff_3 | $(0,0,2,1)$ | 1/6 |
| | | bar_c_star_diff_3 | $(0,0,0,1)$ | 3/160 |
| | | bar_c_star_diff_3 | $(0,0,0,3)$ | 1/6 |
| | | bar_c_star_tau | $(3,0,0,0)$ | 1/2 |
| | | bar_c_star_tau | $(1,2,0,0)$ | -5/6 |
| | | bar_c_star_tau | $(1,0,2,0)$ | -5/6 |
| | | bar_c_star_tau | $(1,0,0,2)$ | -5/6 |
| | | bar_c_star_sigma | $(2,2,0,0)$ | 1/6 |
| | | bar_c_star_sigma | $(2,0,2,0)$ | 1/6 |
| | | bar_c_star_sigma | $(2,0,0,2)$ | 1/6 |
| | | bar_c_star_sigma | $(0,2,0,0)$ | -1/3 |
| | | bar_c_star_sigma | $(0,4,0,0)$ | -1/6 |
| | | bar_c_star_sigma | $(0,2,2,0)$ | -1/3 |
| | | bar_c_star_sigma | $(0,2,0,2)$ | -1/3 |
| | | bar_c_star_sigma | $(0,1,1,1)$ | sqrt(10)/10 |
| | | bar_c_star_sigma | $(0,0,2,0)$ | 1/3 |
| | | bar_c_star_sigma | $(0,0,4,0)$ | -1/6 |
| | | bar_c_star_sigma | $(0,0,2,2)$ | -1/3 |
| | | bar_c_star_sigma | $(0,0,0,2)$ | -3/40 |
| | | bar_c_star_sigma | $(0,0,0,4)$ | -1/6 |

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